A new lower bound for the sixth Schur number: $S(6) \geq 537$

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Abstract

We give an explicit partition of $\{1, \ldots, 537\}$ into six sum-free sets, establishing $S(6) \geq 537$. This improves the lower bound 536 of Fredricksen and Sweet (2000). Known recursive constructions then give improved lower bounds for $S(8)$, $S(11)$, $S(13)$, and $R_6(3)$. We also report 131 pairwise inequivalent partitions found by the search. We describe their structure and provide the complete partitions in the supplementary material.

Keywords: Schur numbers, sum-free partitions, Ramsey theory.

1 Introduction

A set C of integers is *sum-free* if there are no $x, y \in C$, not necessarily distinct, with $x + y \in C$. The Schur number $S(k)$ is the largest n for which $\{1, \ldots, n\}$ can be partitioned into k sum-free sets. The values $S(1), \ldots, S(5) = 1, 4, 13, 44, 160$ are known, the last one determined by Heule [8]. For $k = 6$ the record has been $S(6) \geq 536$ since 2000, when Fredricksen and Sweet [7] found it by a search over symmetric sum-free partitions; it has been quoted as the best known bound ever since. We improve it by one.

Theorem 1. $S(6) \geq 537$.

Proof. The following six sets are pairwise disjoint, their union is $\{1, \ldots, 537\}$, and none of them contains x, y and $x + y$. We list every element, as in the explicit constructions of Fredricksen and Sweet [7]. These properties are verified by checking all sums $x+y$ within each set, including $x = y$.

Color 1 (76 integers).

```
  1   5  12  15  19  21  23  25  29  32  39  43  45  49  95
102 106 108 112 119 122 126 128 130 132 136 139 146 150 152
156 163 166 170 172 174 176 180 358 362 364 366 368 372 375
382 386 388 392 399 402 406 408 410 412 416 419 426 430 432
436 443 489 493 495 499 506 509 513 515 517 519 523 526 533
537
```

Color 2 (76 integers).

```
  2   6   7  17  18  22  27  37  41  42  46  51  66  70  81
 89  90 100 101 105 109 110 113 114 124 125 129 133 134 144
145 149 153 157 158 168 169 192 346 369 370 380 381 385 389
393 394 404 405 409 413 414 424 425 428 429 433 437 438 448
449 457 468 472 487 492 496 497 501 511 516 520 521 531 532
536
```

Color 3 (88 integers).

```
  3   8   9  14  20  24  26  30  31  36  47  53  59  63  65
 69  76  82  86  88  92  98 104 115 120 121 127 131 137 142
143 148 154 159 160 165 171 177 181 182 187 188 193 204 334
345 350 351 356 357 361 367 373 378 379 384 390 395 396 401
407 411 417 418 423 434 440 446 450 452 456 462 469 473 475
479 485 491 502 507 508 512 514 518 524 529 530 535
```

Color 4 (70 integers).

```
  4  10  11  13  16  28  33  34  35  40  55  57  79  93  94
 96  99 111 116 117 118 123 135 138 140 141 147 155 161 162
164 167 179 185 203 335 353 359 371 374 376 377 383 391 397
398 400 403 415 420 421 422 427 439 442 444 445 459 481 483
498 503 504 505 510 522 525 527 528 534
```

Color 5 (76 integers).

```
 38  44  48  50  52  54  56  58  60  61  62  64  67  68  71
 72  73  74  75  77  78  80  83  84  85  87  91  97 103 107
173 189 196 197 201 209 212 236 302 326 329 337 341 342 349
365 431 435 441 447 451 453 454 455 458 460 461 463 464 465
466 467 470 471 474 476 477 478 480 482 484 486 488 490 494
500
```

Color 6 (151 integers).

```
151 175 178 183 184 186 190 191 194 195 198 199 200 202 205
206 207 208 210 211 213 214 215 216 217 218 219 220 221 222
223 224 225 226 227 228 229 230 231 232 233 234 235 237 238
239 240 241 242 243 244 245 246 247 248 249 250 251 252 253
254 255 256 257 258 259 260 261 262 263 264 265 266 267 268
269 270 271 272 273 274 275 276 277 278 279 280 281 282 283
284 285 286 287 288 289 290 291 292 293 294 295 296 297 298
299 300 301 303 304 305 306 307 308 309 310 311 312 313 314
315 316 317 318 319 320 321 322 323 324 325 327 328 330 331
332 333 336 338 339 340 343 344 347 348 352 354 355 360 363
387
```

□

Theorem 1 needs only this one partition. Section 2 lists what it implies for other Schur and Ramsey numbers. In Section 3 we describe all 131 partitions of $\{1, \ldots, 537\}$ into six sum-free sets that our search produced. Section 4 describes the construction of the families. The complete partitions and supplementary tables are available at `https://github.com/jeffchen006/Schur-Number-6`.

Throughout, $N = 538$, $\mathrm{f}(x) = \min(x, N - x)$, and a colouring c of $\{1, \ldots, 537\}$ is *palindromic* if $c(x) = c(N - x)$ for all x. We number the colours $1, \ldots, 6$ in the order of their least elements.

2 Consequences for other bounds

Classical recurrences of Schur and Abbott–Hanson [1, 2], and the template constructions of Rowley and Ageron et al. [3, 4], turn smaller partitions into larger ones. We use

$$S(k+2) \geq 10S(k) + 2 \quad [5], \qquad S(k+5) \geq 380S(k) + 148 \quad [4].$$

A sum-free k-partition of $[1, n]$ also gives an edge coloring of K_{n+1} without monochromatic triangles: color ij by the class of $|i - j|$. Consequently, $R_k(3) \geq S(k) + 2$.

Corollary 2. *The lower bounds in the column headed "This work" in Table 1 hold.*

Table 1: Lower bounds in the cited literature and the bounds obtained here. The six- and eight-color bounds are supported by explicit partitions.

Quantity	Previous bound	This work	Previous source
$S(6)$	536	537	Fredricksen–Sweet [7]
$S(8)$	5 362	5 372	Bengone et al. [5]
$S(11)$	203 828	204 208	Ageron et al. [4]
$S(13)$	2 038 282	2 042 082	Bengone et al. [5]
$R_6(3)$	538	539	Fredricksen–Sweet [7, 6]

Proof. $S(8) \geq 10 \cdot 537 + 2$, $S(11) \geq 380 \cdot 537 + 148$, $S(13) \geq 10 \cdot S(11) + 2 \geq 10 \cdot 204\,208 + 2$, and $R_6(3) \geq S(6) + 2$. For $S(8)$ we also built the partition explicitly: apply the width-10 template of [5, Table II] to the partition of Theorem 1, with colours 7 and 8 for the labels A and B. This gives a partition of $\{1, \ldots, 5372\}$ into eight sum-free sets. □

Bounds for larger k that are derived through these recurrences change in the same way; we list only the four cases in which we checked the previously published values.

3 The 537-partitions and their families

The example above is palindromic, and each of its classes is even sum-free modulo 538. The least elements of its classes are $1, 2, 3, 4, 38, 151$, and the sixth class contains the whole interval $[237, 301]$. These features are captured by the following family of colourings.

Definition 3. A colouring $c\colon \{1, \ldots, 537\} \to \{1, \ldots, 6\}$ whose classes are sum-free modulo 538 is in *frame* g if, with $Q = 151$ and $d = 38$,

$$\begin{aligned}
\mathrm{f}(x) < j &\;\Rightarrow\; c(x) \neq j \quad (j = 1, 2, 3, 4),\\
\mathrm{f}(x) < d \text{ or } Q - d < \mathrm{f}(x) < Q &\;\Rightarrow\; c(x) \neq 5,\\
\mathrm{f}(x) \in \{d,\, N - 2Q\} &\;\Rightarrow\; c(x) = 5,\\
\mathrm{f}(x) < Q &\;\Rightarrow\; c(x) \neq 6,\\
\mathrm{f}(x) = Q \text{ or } \mathrm{f}(x) > N - 2Q &\;\Rightarrow\; c(x) = 6.
\end{aligned}$$

In words: the sixth colour does not occur below 151 and fills the middle of the interval, and the fifth colour does not occur below 38 or just below 151. Figure 1 shows the certificate and these restrictions.

The first class of the example has a simple description as well. Put

$$\begin{aligned}
A &= \{1, 5, 12, 15, 19, 21, 23, 25, 29, 32, 39, 43, 45, 49\},\\
B &= A \cup (151 - A) \cup \{151 + a : a \in A,\ a \leq 29\} \cup \{95\},
\end{aligned}$$

a set of 38 numbers, and $C^* = \{x \in \{1, \ldots, 537\} : \mathrm{f}(x) \in B\}$, a set of 76 numbers. The first class of the example is exactly C^*.

Proposition 4. *The* 131 *partitions of* $\{1, \ldots, 537\}$ *into six sum-free sets listed in the supplementary material are pairwise distinct, every class of each of them is sum-free modulo* 538*, and no one of them is obtained from another by a map* $x \mapsto rx \bmod 538$ *with* $\gcd(r, 538) = 1$ *followed by renaming the colours. They form the five families summarized in Table 2. The example of Theorem 1 is* $\mathcal{F}_1.001$*. Each partition in* $\mathcal{F}_1$*–*$\mathcal{F}_4$ *agrees with the example in at least* 462 *of the* 537 *positions.*

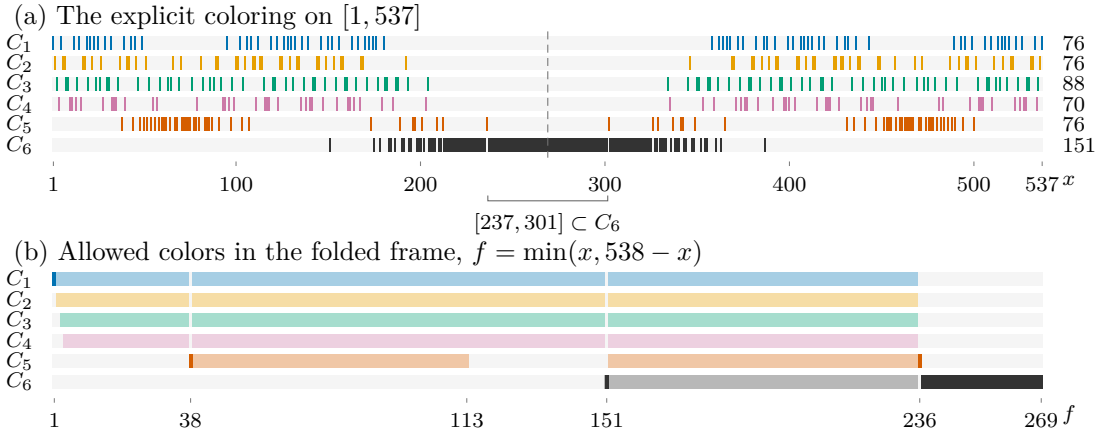


Figure 1: The 537 certificate and its search frame. (a) A colored cell in row C_j marks an integer in that class; the right margin gives the class size. The dashed line marks the center of reflection. (b) Light cells mark permitted colors before propagation; dark cells mark forced colors. The row layout follows the certificate diagrams of Heule [8, Fig. 1]; the entries here are generated from our certificate and frame rules.

Table 2: The 131 retained partitions. Intervals give the observed minimum and maximum within a family. Features of $\mathcal{F}_5$ use the normalization stated in the text. All entries are computed from the 131 explicit partitions.

Family	Count	d_{pal}	Δ_1	h
$\mathcal{F}_1$	107	0	0	0
$\mathcal{F}_2$	14	0	2	0
$\mathcal{F}_3$	5	2–12	0	0
$\mathcal{F}_4$	4	0	2–8	1–13
$\mathcal{F}_5$	1	5	3	5
Total	131			

Proof. For each partition, check all sums within each class modulo 538. To check inequivalence, compare all images under multiplication by units modulo 538, with colors renumbered in the order of their least elements. The family properties follow by counting the quantities in Table 2. □

For Table 2, d_{pal} counts pairs $\{x, 538-x\}$ with different colors, $\Delta_1 = |C_1 \triangle C^*|$, and h counts folded positions $f > 236$ outside color 6. Family $\mathcal{F}_5$ is measured after $x \mapsto 43x \bmod 538$ and color renumbering; its original class minima are $1, 2, 3, 4, 7, 16$. Families $\mathcal{F}_1$–$\mathcal{F}_3$ satisfy frame g. Family $\mathcal{F}_4$ satisfies the frame with the solid-middle condition removed. These families describe the retained sample, not all possible 537-partitions.

4 Construction of the families

The palindromic families are obtained by imposing reflection symmetry and the sum-free conditions modulo 538. For $\mathcal{F}_1$ and $\mathcal{F}_2$, we also impose frame g. The two families are distinguished by whether the first class is C^* or differs from it in one reflected pair. For $\mathcal{F}_3$, we retain the frame conditions but remove reflection symmetry. For $\mathcal{F}_4$, we retain reflection symmetry but remove the condition that color 6 contains every position with $f(x) > 236$.

For $\mathcal{F}_5$, let c be the coloring in Theorem 1 and put

$$c_m(x) = c(mx \bmod 538), \qquad m \in \{3, 25, 221\},$$

with the colors of each c_m renamed to maximize agreement with c. We then require a new coloring c' to satisfy

$$c'(x) \in \{c_3(x), c_{25}(x), c_{221}(x)\}, \qquad 1 \le x \le 537,$$

together with sum-freeness modulo 538. This construction does not impose reflection symmetry or the solid-middle condition.

The example in Theorem 1 is sufficient for the new lower bound. The other partitions describe variations of its structure; they do not form an exhaustive classification of six-color partitions of $[1, 537]$.

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